C.D.U./U.D.C. : [621.316.57.027.2+621.316.933.027.2] 001.2.001.4(083.71) 65.014.13

14

COMMISSION ÉLECTROTECHNIQUE INTERNATIONALE

NORME DE LA CEI

INTERNATIONAL ELECTROTECHNICAL COMMISSION

IEC STANDARD

Publication 157-1B

1979

Deuxième complément à la Publication 157-1 (1973)

Appareillage à basse tension

Première partie: Disjoncteurs

Second supplement to Publication 157-1 (1973)

Low-voltage switchgear and controlgear

Part 1: Circuit-breakers

Descripteurs: disjoncteurs à basse tension ; fusibles à basse tension ; exigences ; essais ; définitions.

Descriptors: low-voltage circuit-breakers; low-voltage fuses; requirements; testing; definitions.

Droits de reproduction réservés — Copyright - all rights reserved

Aucune partie de cette publication ne peut être reproduite ni utilisée sous quelque forme que ce soit et par aucun procédé, électronique ou mécanique, y compris la photocopie et les microfilms, sans l'accord écrit de l'éditeur.

No part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from the publisher.

Bureau Central de la Commission Electrotechnique Internationale

1, rue de Varembé

Genève, Suisse

Prix
Price Fr. s. 32.–

IEC
VOL.9

INTERNATIONAL ELECTROTECHNICAL COMMISSION

Second supplement to Publication 157-1 (1973)
LOW-VOLTAGE SWITCHGEAR AND CONTROLGEAR
Part 1: Circuit-breakers

FOREWORD

1) The formal decisions or agreements of the IEC on technical matters, prepared by Technical Committees on which all the National Committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.

2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.

3) In order to promote international unification, the IEC expresses the wish that all National Committees should adopt the text of the IEC recommendation for their national rules in so far as national conditions will permit. Any divergence between the IEC recommendation and the corresponding national rules should, as far as possible, be clearly indicated in the latter.

PREFACE

This standard has been prepared by Sub-Committee 17B: Low-voltage Switchgear and Controlgear, of IEC Technical Committee No. 17: Switchgear and Controlgear.

It combines two documents prepared by two different working groups.

Elaboration of Part A: Amendments to be made to IEC Publications 157-1 (2nd edition, 1973) and 157-1A (1976)

New Clauses 4.3.2.1, 4.3.2.2 and 8.2.2 are the object of the first document for which a first draft was circulated in August 1975, according to the decision taken in Paris in April 1974. A second draft was circulated in December 1976 and discussed in Moscow in June 1977. The draft, Document 17B(Central Office)98, was submitted to the National Committees for approval under the Six Months' Rule in January 1978.

The following countries voted explicitly in favour of publication:

Australia	Japan
Belgium	Netherlands
Bulgaria	Poland
Canada	South Africa (Republic of)
China	Spain
Egypt	Sweden
Finland	Turkey
France	Union of Soviet Socialist Republics
Germany	United Kingdom
Israel	United States of America

Other Clauses of Part A are the object of the second document for which a first draft was circulated in July 1973, according to the decision taken in Stockholm in September 1972, and discussed in Paris in February 1974.

A second draft was circulated in September 1974, according to the decision taken in Paris and discussed in The Hague in September 1975. A third draft was circulated in December 1975, according to the decision taken in The Hague in December 1975. A fourth draft was circulated in October 1976 and discussed in Moscow in June 1977. The draft, Document 17B(Central Office)99, was submitted to the National Committees for approval under the Six Months' Rule in January 1978.

The following countries voted explicitly in favour of publication:

- Australia
- Belgium
- Bulgaria
- Canada
- Egypt
- Finland
- France
- Germany
- Hungary
- Japan
- Netherlands
- Poland
- South Africa (Republic of)
- Spain
- Sweden
- Turkey
- Union of Soviet Socialist Republics
- United Kingdom
- United States of America

Elaboration of Part B: Appendix D:
Co-ordination of circuit-breakers with separate fuses associated in the same circuit

Part B comprises Appendix D to Publication 157-1, as included in Publication 157-1A (1976) and amended in compliance with the above Document 17B(Central Office)99.

Other IEC publications quoted in this standard:

Publications 157-1A: First supplement to Publication 157-1 (1973): "Low-voltage Switchgear and Controlgear, Part 1: Circuit-breakers.

157-2: (Under consideration).

269-1: Low-voltage Fuses. Part 1: General requirements.

269-2: Part 2: Supplementary Requirements for Fuses for Industrial Applications.

269-3: Part 3: Supplementary Requirements for Fuses for Domestic and Similar Applications.

439: Factory-built Assemblies of Low-voltage Switchgear and Controlgear.

Second supplement to Publication 157-1 (1973)
LOW-VOLTAGE SWITCHGEAR AND CONTROLGEAR
Part 1: Circuit-breakers

A — AMENDMENTS TO BE MADE TO PUBLICATIONS 157-1 (2nd EDITION, 1973) AND 157-1A (1976)

Amendment to Clause 1.1

Reword fourth paragraph to read:

"The requirements for circuit-breakers intended to be accessible to and operated by unskilled people on domestic and similar installations are contained in Publication 157-2 (under consideration).

Particular requirements for circuit-breakers which are also intended to provide earth-leakage protection are under consideration."

Amendment to Clause 2.2.8

Amend this clause to read:

"The portion of a circuit-breaker associated exclusively with one electrically separated conducting path of its main circuit, provided with contacts intended to connect and disconnect the main circuit itself, and excluding those portions which provide a means for mounting and operating all poles together."

Amendment to Clause 2.5.9

Add the following note:

"*Note.* — Account has to be taken both of the I^2t let-through and also of the electro-mechanical effects associated with the peak current."

Amendment to Clause 2.5.21

Reword this clause as follows:

"The value of current corresponding to the intersection of the characteristics of operation of two over-current protecting devices (either I^2t or homogeneous time-current characteristic curves).

Note. — When a circuit-breaker is associated with a back-up device (fuse or another circuit-breaker) generally intended for protecting it from short-circuits exceeding its breaking capacity, one must distinguish:

a) the take-over current I_B: the maximum over-current value beyond which the back-up device will necessarily operate;

Note. — The take-over current may also be defined as the back-up limit of current.

b) the selectivity limit of current I_S: the maximum over-current value at which the circuit-breaker will operate alone (see Figures 7 and 8, page 21, which shall replace Figure A of Publication 157-1A)."

Insert the following new definitions:

"2.5.26 *Joule integral (I^2t)*

The integral of the square of the current over a given time interval:

$$I^2t = \int_{t_0}^{t_1} I^2 dt$$

Note. — This definition is that of the International Electrotechnical Vocabulary (I.E.V.) (441-07-14) without the notes.

"2.5.26.1 *I^2t characteristic of a circuit-breaker*

A curve giving the values of I^2t related to break times (generally minimum or maximum values of I^2t) as a function of the prospective current under stated conditions of operation.

Note. — Normally it is sufficient to quote maximum and minimum values of I^2t at rated breaking capacity.

"2.5.26.2 *I^2t zone of a circuit-breaker*

The zone contained between the minimum I^2t and the maximum I^2t characteristics."

4.1 and 4.3

Replace "Rated quantities" by "Rated values".

Replace Clause 4.3.2.1 by the two new clauses, as follows:

"4.3.2.1 *Rated conventional thermal current*

The rated conventional thermal current (I_{th}) of a circuit-breaker is the maximum current stated by the manufacturer that the unenclosed circuit-breaker can carry in 8 hour duty (see Clause 4.3.4.1) when tested in free air, without the temperature rise of its several parts exceeding the limits specified in Clause 7.3 (Tables IV and V) when tested according to Clause 8.2.2.

Notes 1. — Free air is understood to be that obtained under normal indoor conditions reasonably free from draughts and external radiation.

2. — An unenclosed circuit-breaker is a circuit-breaker supplied by the manufacturer without an enclosure or a circuit-breaker supplied by the manufacturer with an enclosure forming an integral part of the circuit-breaker.

"4.3.2.2 *Rated enclosed thermal current*

The rated enclosed thermal current (I_{the}) of a circuit-breaker is the maximum current stated by the manufacturer that the circuit-breaker can carry in the stated duty (see Clause 4.3.4) when mounted in a specified enclosure. Tests for this rating shall be in accordance with Clause 8.2.2, but are not mandatory if the test for "rated conventional thermal current" has been made, and the manufacturer is prepared to state an enclosed thermal current rating.

The rating may be an unventilated rating, in which case the enclosure shall be of the size stated by the manufacturer to be the smallest enclosure that is applicable in service. Alternatively, the rating may be a ventilated rating with the ventilation in accordance with the manufacturer's data.

Note. — It is not possible to usefully define a rated service thermal current as the installation and service conditions can vary greatly. (The "rated current" Sub-clause 4.2 of IEC Publication 439 is in effect the "rated service thermal current".)".

Renumber present Clause 4.3.2.2 as 4.3.2.3.

Replace Clauses 8.2.2.1 and 8.2.2.2 by the following clauses:

"8.2.2.1 *Ambient air temperature*

The ambient air temperature shall be measured during the last quarter of the test period by means of at least two thermometers or thermocouples equally distributed around the circuit-breaker at about half its height and at a distance of about 1 m from it. The thermometers or thermocouples shall be protected against air currents, heat radiation and indicating errors due to rapid temperature changes.

"8.2.2.2 *Temperature-rise tests of the main circuit*

The circuit-breaker shall be mounted approximately as under usual service conditions and shall be protected against undue external heating or cooling.

Circuit-breakers having an integral enclosure and circuit-breakers only intended for use with a special type of enclosure shall be tested in their enclosure for the rated conventional thermal current test. No opening giving false ventilation shall be allowed.

Details of any enclosure, ventilation arrangements, and sizes of test conductors shall be stated in the test report.

For tests with a.c. single-phase or d.c. currents, the test current shall be not less than the rated conventional thermal current. For tests with multi-phase currents, the current shall be balanced in each phase within ±5%, and the average of these currents shall be not less than the rated conventional thermal current.

The temperature-rise test of the main circuit is made at the rated conventional thermal current.

Tests on d.c. rated circuit-breakers may be made with an a.c. supply for convenience of testing, but only with the consent of the manufacturer. Tests on a.c. rated circuit-breakers shall be made at a frequency of between 45 Hz and 62 Hz where the rated frequency of the circuit-breaker is 50 Hz or 60 Hz; for lower or higher rated frequencies, a tolerance of ±20% shall apply.

The test shall be made over a period of time sufficient for the temperature rise to reach a steady-state value, but not exceeding 8 h. In practice, this condition is reached when the variation does not exceed 1 degree Celsius per hour.

Note. — In practice, to shorten the test the current may be increased during the first part of the test, it being reduced to the specified test current afterwards.

At the end of the test, the temperature rise of the different parts of the main circuit shall not exceed the values specified in Table V.

Depending on the value of the rated thermal current, one of the following procedures shall be adopted:

For values of test current up to and including 400 A:

a) The connections shall be single-core, PVC insulated, copper cables or wires with cross-section areas as given Table IX.

b) In the case of multi-pole circuit-breakers, tested with a.c., the test may be carried out with single-phase current with all poles connected in series provided magnetic effects can be neglected.

c) The connections shall be in free air, and spaced at approximately the distance existing between the terminals.

d) For single-phase or multi-phase tests, the minimum length of any temporary connection from a circuit-breaker terminal to another terminal or to the test supply or to a star point shall be:

— 1 m for cross-sections up to and including 35 mm²;
— 2 m for cross-sections larger than 35 mm² (or AWG 2).

For values of test current higher than 400 A but not exceeding 800 A:

a) The connections shall be single core, PVC insulated, copper cables with cross-section areas as given in new Table X, or the equivalent copper bars given in Table X as recommended by the manufacturer.

b) In the case of multi-pole circuit-breakers, tested with a.c., the test may be carried out with single-phase current with all poles connected in series provided magnetic effects can be neglected.

c) Cables or copper bars shall be spaced at approximately the distance between terminals. Copper bars shall be finished matt black. Multiple parallel cables per terminal shall be bunched together and arranged with approximately 10 mm air space between each other. Multiple copper bars per terminal shall be spaced at a distance approximately equal to the bar thickness. If the sizes stated for the bars are not suitable for the terminals, or are not available, it is allowed to use other bars having approximately the same cross-section and approximately the same or smaller cooling surface. Cables or copper bars shall not be interleaved.

d) For single-phase or multi-phase tests, the minimum length of any temporary connection from a circuit-breaker terminal to another terminal or to the test supply shall be 2 m. The minimum length to a star point may be reduced to 1.2 m.

For values of test current higher than 800 A but not exceeding 3150 A:

a) The connections shall be copper bars of the sizes stated in Table X unless the circuit-breaker is designed only for cable connection. In this case, the size and arrangement of the cables shall be as specified by the manufacturer.

b) In the case of multi-pole circuit-breakers, tested with a.c., the test may be carried out with single-phase current with all poles connected in series provided magnetic effects can be neglected.

c) Copper bars shall be spaced at approximately the distance between terminals. Copper bars shall be finished matt black. Multiple copper bars per terminal shall be spaced at a distance approximately equal to the bar thickness. If the sizes stated for the bars are not suitable for the terminals, or are not available, it is allowed to use other bars having approximately the same cross-section and approximately the same or smaller cooling surfaces. Copper bars shall not be interleaved.

d) For single-phase or multi-phase tests, the minimum length of any temporary connection from a circuit-breaker terminal to another circuit-breaker or to the test supply shall be 3 m, but this can be reduced to 2 m provided that the temperature rise at the supply end of the connection is not more than 5 °C below the temperature rise in the middle of the connection length. The minimum length to a star point shall be 2 m.

For values of test current higher than 3150 A:

Agreement shall be reached between manufacturer and user on all relevant items of the test, such as: type of supply, number of phases and frequency (where applicable), cross-sections of test connections, etc. This information shall form part of the test report.

Note. — In all cases, the use of single-phase a.c. current for testing multi-phase circuit-breakers is only permissible if magnetic effects are small enough to be neglected. This requires careful consideration especially for currents above 400 A.".

Replace Table X, page 63, by the new Table X, as follows:

Table X

Standard test conductors for rated conventional thermal currents higher than 400 A

Value of rated conventional thermal current (A)	Range of rated conventional thermal current (A)	Test connection			
		Cables		Copper bars	
		Quantity	Cross-sections in mm²	Quantity	Dimensions in mm
500	400 - 500	2	150 (16)	2	30× 5 (15)
630	500 - 630	2	185 (18)	2	40× 5 (15)
800	630 - 800	2	240 (21)	2	50× 5 (17)
1 000	800 - 1 000		—	2	60× 5 (19)
1 250	1 000 - 1 250		—	2	80× 5 (20)
1 600	1 250 - 1 600		—	2	100× 5 (23)
2 000	1 600 - 2 000		—	3	100× 5 (20)
2 500	2 000 - 2 500		—	4	100× 5 (21)
3 150	2 500 - 3 150		—	3	100×10 (23)

Notes 1. — Value of current shall be greater than the first value and less than or equal to the second value.

2. — Bars are assumed to be arranged with their long faces vertical. Arrangements with long faces horizontal may be used if specified by the manufacturer.

3. — Values in brackets are estimated temperature rises of the test conductors given for reference.

8.2.4.3 *Standard tests*

Add at the end of this clause:

"The maximum value of I^2t (see clause 2.5.26) noted during these tests shall be recorded on the test report.

Note. — The maximum value of I^2t recorded during the tests may not be the maximum possible value for the prescribed conditions. Additional tests are necessary for determining this maximum value."

9.5.1 *Verification of the take-over current*

Second line: replace "(see Figure A, page 18)" by "(see Figures 7 and 8, page 21)".
Add the following new clause:

"10 *Additional requirements for four-pole circuit-breakers*

"10.1 The fourth pole shall only be intended for connecting the neutral, and shall be clearly marked to that effect by the letter N.

"10.2 The fourth pole shall operate together with the other poles.

"10.3 The fourth pole may be fitted with an over-current release.

"10.4 For a circuit-breaker having a value of rated thermal current not exceeding 200 A, the thermal rating of all four poles shall be identical.

For higher thermal current ratings, the fourth pole may have a rating not less than 50% of that of each of the other poles but not less than 200 A.

"10.5 A temperature rise test in accordance with Clause 8.2.2 shall first be made on the three poles which incorporate over-current releases.

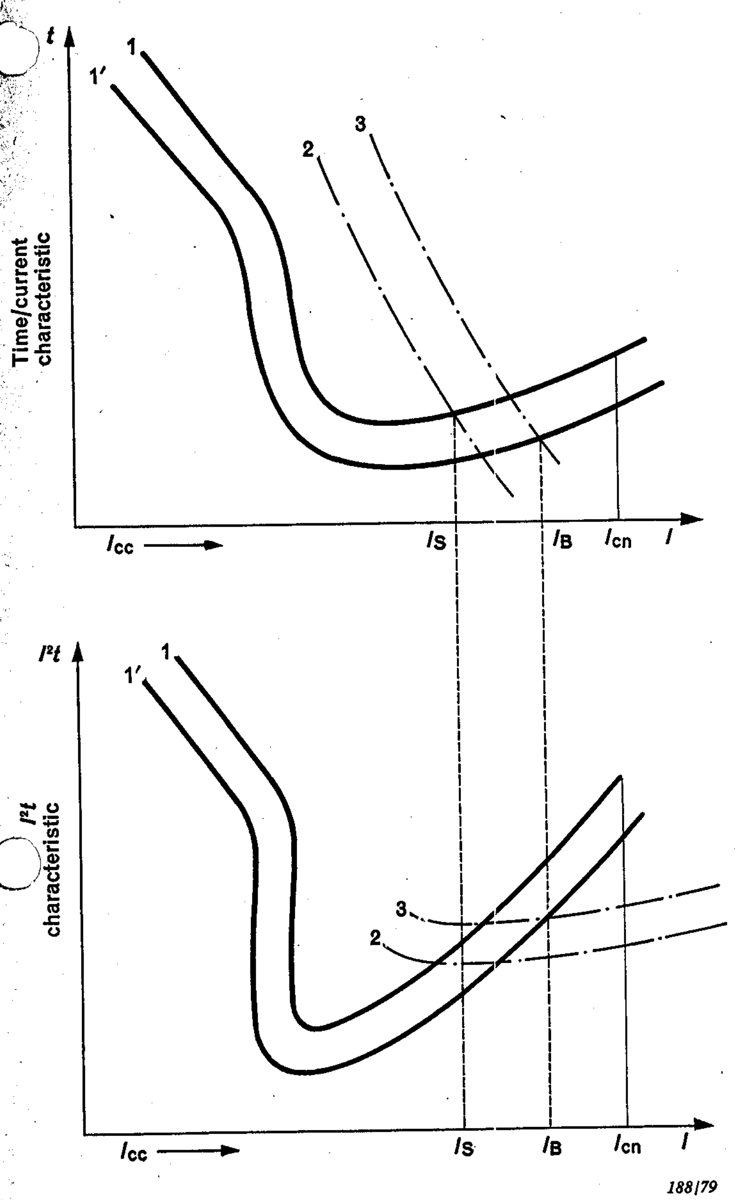
For a circuit-breaker having a value of rated thermal current not exceeding 200 A, a separate and additional test shall be made by passing the test current through the fourth pole and its adjacent pole.

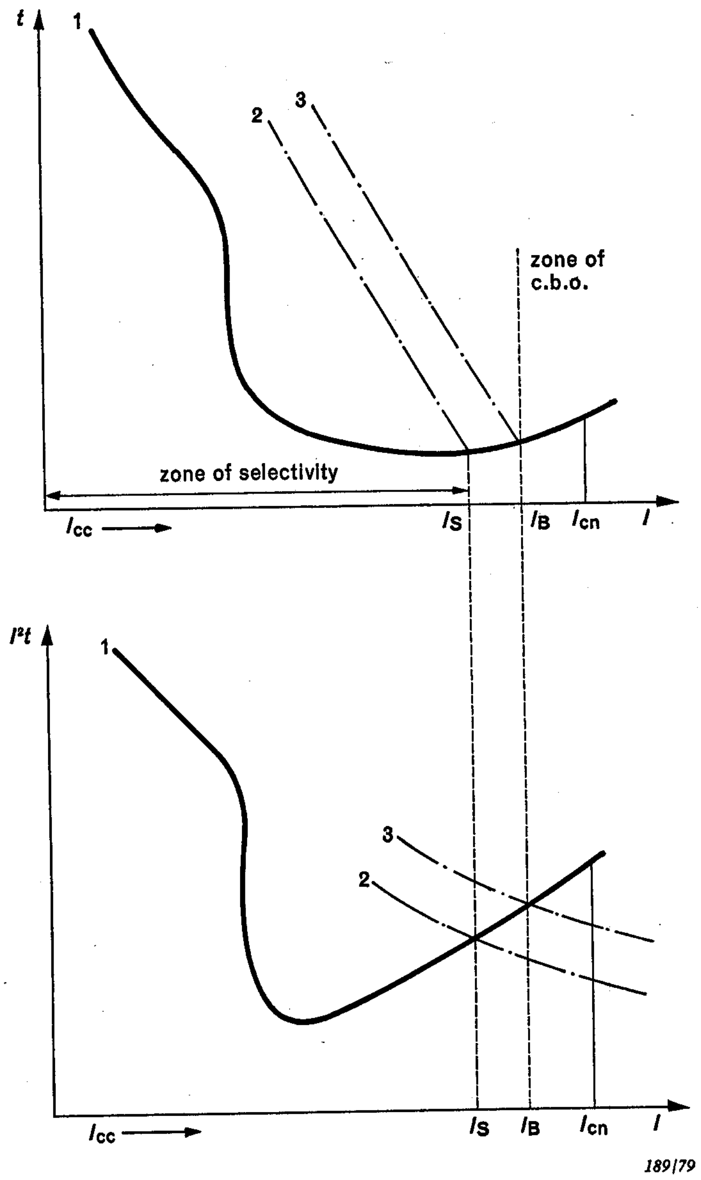
For higher thermal current ratings, the method of testing shall be the subject of a separate agreement between manufacturer and user.

"10.6 For short-circuit tests in accordance with Clause 8.2.4, the neutral point on the load side of the circuit-breaker on test shall be bonded back to the neutral of the supply (or to its artificial neutral) (see Figure 1).

"10.7 An additional short-circuit test in accordance with Clause 8.2.4 (see Publications 157-1 and 157-1A) shall be made on the fourth pole and its adjacent pole at rated breaking capacity, using the circuit shown in Figure 2."

Replace Figure A of Publication 157-1A by the new Figures 7 and 8.





I_{cc} = Prospective short-circuit current;
I_{cn} = Rated short-circuit breaking capacity (see Clause 4.3.5.2);
I_S = Selectivity limit of current (see new Clause 2.5.21*b*));
I_B = Take-over current (see new Clause 2.5.21*a*));
I^2t = The integral of the square of the current extended over the break-time;
1 = Upper limit of the operating range of the circuit-breaker;
1' = Lower limit of the operating range of the circuit-breaker;
2 = Pre-arcing characteristic of the fuse;
3 = Operating characteristic of the fuse;
Zone of c.b.o. = zone of certain back-up operation.

FIG. 7. — General case of circuit-breakers with break-time exceeding 40 ms or 50 ms (in particular circuit-breakers fitted with time-lag release independent from overcurrent).

FIG. 8. — General case of circuit-breakers with break-time less than 20 ms (in particular current-limiting circuit-breakers).

B — APPENDIX D: CO-ORDINATION OF CIRCUIT-BREAKERS WITH SEPARATE FUSES ASSOCIATED IN THE SAME CIRCUIT

Note. — This appendix supersedes Appendix D of Publication 157-1A (1976).

Introduction

With integrally-fused circuit-breakers, the circuit-breaker manufacturer is responsible for ensuring the correct co-ordination of the circuit-breaker with its fuses. But circuit-breakers are frequently connected in series with separate fuses, for reasons such as the method of power distribution adopted for the installation, or because the breaking capacity of the circuit-breaker alone may be insufficient for the proposed application. In such instances, the fuses may be mounted in locations remote from the circuit-breaker. Moreover, the fuse or fuses may be protecting a main feeder supplying a number of circuit-breakers or just an individual circuit-breaker.

Note. — Generally, fuses are located on the supply side of the circuit-breaker in such an association. However, specific installation rules may authorize to locate them on the load side of the circuit-breaker. As regards co-ordination, problems are the same in both cases.

For such applications, the user or specifying authority may have to decide, on the basis of a desk study alone, how best the optimum level of co-ordination may be achieved. This appendix is therefore intended to give guidance on this decision, and also to give guidance on the type of information the circuit-breaker manufacturer should make available to the prospective user.

Note. — The term "co-ordination" includes consideration of discrimination (i.e. selective operation) as well as consideration of back-up protection (see new Clause 2.5.21).

Guidance is also given on test requirements, should such tests be deemed essential for the proposed application. But for the vast majority of applications, expensive and complicated testing may not be considered necessary. This may be for a variety of practical considerations, such as the prospective short-circuit current being less than or only marginally in excess of the breaking capacity of the circuit-breaker alone.

D1. Scope

This appendix deals with requirements for the co-ordination of circuit-breakers with separate fuses associated in the same circuit.

D2. Object

The object of this appendix is to state:

- the general requirements for the co-ordination of a circuit-breaker with its associated fuse or fuses;
- the methods and the tests (if deemed necessary) intended to verify that the conditions for co-ordination have been met.

D3. General requirements for co-ordination of a circuit-breaker with its associated fuse or fuses

D3.1 *General considerations*

Ideally, the co-ordination should be such that the circuit-breaker alone will operate at all values of over-current up to the limit of its rated breaking capacity.

In practice, the following considerations apply:

a) If the value of the prospective fault current at the point of installation is less than the rated breaking capacity of the circuit-breaker, it may be assumed that the fuse or fuses are only in the circuit for considerations other than those of back-up protection. If the value of the take-over current I_B (see definition 2.5.21) is too low, there is a risk of unnecessary loss of selectivity (i.e. selective operation).

b) If the value of the prospective fault current at the point of installation exceeds the rated breaking capacity of the circuit-breaker, the fuse or fuses shall be so selected as to ensure compliance with the requirements of Clauses D3.2 and D3.3.

D3.2 *Take-over current*

The take-over current I_B (see definition 2.5.21) shall be not greater than the breaking capacity of the circuit-breaker alone.

D3.3 *Behaviour of the circuit-breaker in association with its fuses*

For all values of over-current up to and including the breaking capacity assigned to the combination:

a) The making operation of the circuit-breaker as well as the breaking operation of the combination shall not give rise to external manifestations (such as emission of flames) projected beyond the limits stated by the manufacturer (see Clause 9.2.2*a)* in Publications 157-1 and 157-1A).

b) There shall be no flashover between poles or between poles and frame, nor welding of contacts (see Clause 9.2.2*b)* in Publication 157-1).

Note. — See also Clauses D5.2 and D5.3.

D4. Type and characteristics of the associated fuses

On request, the manufacturer of the circuit-breaker shall state, in accordance with the relevant IEC publications (Publications 269-1 and 269-2 or 269-3), the type and the characteristics of the fuses to be used with the circuit-breaker, and the maximum prospective short-circuit current for which the combination is suitable at the stated rated operational voltage.

Whenever possible, the fuses shall be located on the supply side of the circuit-breaker. If the fuses are located on the load side, it is essential that the connections between the circuit-breaker and the fuses be so designed as to minimize any risk of short circuit.

D5. Methods for verification of the co-ordination

D5.1 *Determination of the take-over current*

Compliance with the requirements of Clause D3.2 shall be verified by comparing the operating characteristics of the circuit-breaker and of the fuse.

If the circuit-breaker is fitted with adjustable over-current opening releases, the operating characteristics to be used shall be those corresponding to the minimum current setting. If the circuit-breaker can be fitted with instantaneous over-current opening releases, the operating characteristics to be used shall be those corresponding to the circuit-breaker fitted with such releases.

D5.2 *Verification of the behaviour of the circuit-breaker/fuse combination under short-circuit conditions*

a) Full compliance with the requirements of Clause D3.3 can be verified only by tests in accordance with Clause D5.3. In this case, all the conditions for the test shall be as specified in Clause 8.2

of this standard, with the adjustable resistors and inductors for the short-circuit test on the supply side of the association.

b) If the circuit-breaker is of the non-current-limiting type, then in some practical cases it may be sufficient to compare the operating characteristics of the circuit-breaker and the fuse, special attention being paid to the following:

1) The Joule-integral values of the circuit-breaker and of the fuse during the total clearing time;

2) The effects on the circuit-breaker (e.g. by arc energy, by maximum peak current, etc.) at the peak value of the cut-off current of the fuse.

The suitability of the association may be evaluated by considering the maximum total operating I^2t value of the fuse, over the range from the rated short-circuit capacity of the circuit-breaker to the prospective short-circuit current of the proposed application, but not exceeding the short-circuit capacity of the association. This value shall not exceed the maximum let-through I^2t of the circuit-breaker at its rated short-circuit breaking capacity or other limiting value stated by the manufacturer.

In the ultimate, only the appropriate tests can verify thoroughly the suitability of the association of fuse and circuit-breaker at all currents up to the limiting value of the breaking capacity of the fuse.

D5.3 *Current for verification of current co-ordination under short-circuit conditions*

The short-circuit test is made with the maximum prospective current I_A, for the proposed application. This shall not exceed the maximum prospective short-circuit current assigned by the manufacturer to the association.

In addition, if I_B is close to the rated breaking capacity (I_{cn}) of the circuit-breaker, for example greater than 80% of I_{cn}, an additional series of tests shall be made at a value of prospective current equal to 120% of I_{cn}. At least one fuse must operate.

Such additional tests may be made on a new and clean circuit-breaker at the manufacturer's request.

If tests are made in accordance with Clause D5.2, the following procedure shall apply:

Order of tests	Test sub-clauses applicable to circuit-breakers of short-circuit performance categories	
	P-1	P-2
Short-circuit test at current I_A	CO 8.2.4	CO 8.2.4
Conditions after test	8.2.4.10 and 8.2.4.10.1	8.2.4.10 and 8.2.4.10.2

D5.4 *Results to be obtained*

See Clause D3.3.